

Dependence-aware evaluation of cyclic-peptide permeability

1 Outcome-blind curation

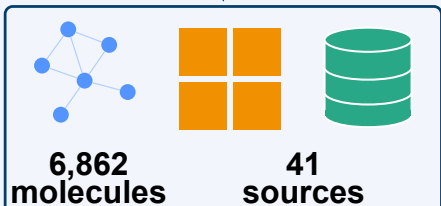
Raw PAMPA compilation
7,298 rows

Exclude 372 censored

6,926 uncensored
usable rows

Collapse 31 compatible
source-structure groups

6,895 records



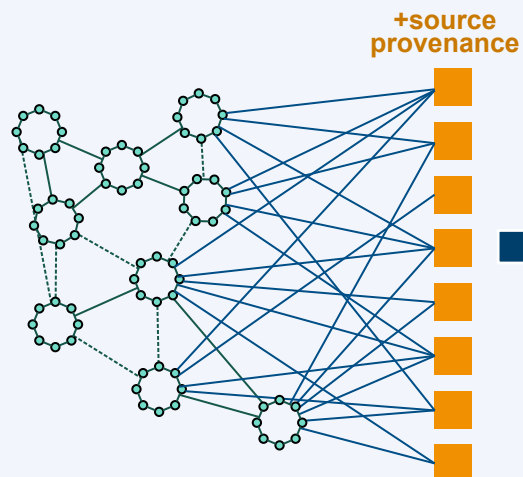
2 Joint dependence blocks

Analogue graph: 141,425 edges

chiral ECFP4 Tanimoto ≥ 0.80

MW ratio 0.80–1.25

exact one-token cyclic edit



305 analogue components

18 indivisible joint blocks

Largest: 4,010 (58.16%)

Top 3: 6,370 (92.39%)



Outcomes never used
to build blocks

3 Nested evaluation

17 training blocks +
1 unseen test block $\times 18$



Training-only inner CV: 4 grouped baskets
impute · scale · filter · select · stop · calibrate



Refit on 17 blocks

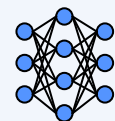
seal prediction

score test block

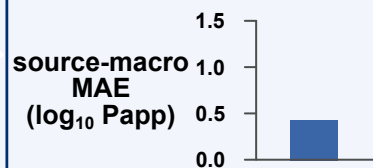
5 seeds: 0, 123, 492, 1107, 1968

4 Same DMPNN, different boundary

DMPNN

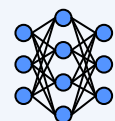


Molecule-grouped 5-fold
MAE 0.4672

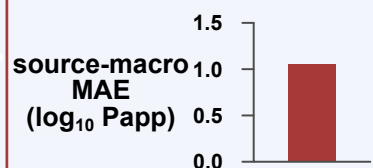


VS.

DMPNN



Joint leave-one-block-out
MAE 1.0046



$\Delta +0.5373$

95% CI
0.3244-0.7530



Fixed D3 = balanced loss + descriptors

MAE 0.9969

Nominal 90%
coverage: 77.0%

H2: 0/5
conditions passed

Evaluation boundary changed the estimated transfer error



Protocol divergence, not the isolated
causal effect of dependence removal.